

I A warehouse has throughput dynamics described by

$$3\dot{y}(t) + y(t) = 6u(t)$$

a) Take the Laplace transform and determine the transfer function

$$G(s) = \frac{Y(s)}{R(s)}$$

1) Set any initial conditions to zero

2) Swap derivatives for s counterparts

$$\Rightarrow 3s Y(s) + Y(s) = 6 R(s)$$

3) Rearrange to get $G(s)$

$$\Rightarrow Y(s)(3s + 1) = 6 R(s)$$

$$\Rightarrow \frac{Y(s)}{R(s)} = \frac{6}{3s + 1}$$

b) Determine the static gain K and time constant τ .

$\Rightarrow$ The TF is a standard 1st-order system

$$\frac{K}{\tau s + 1}$$

$$K = 6, \quad \tau = 3$$

c) If the throughput command is

$$r(t) = 2u(t),$$

determine the long-run value y_{∞} .

$$R(s) = \frac{2}{s}$$

$$Y(s) = G(s)R(s) = \frac{6}{3s+1} \cdot \frac{2}{s}$$

$$Y(s) = \frac{12}{s(3s+1)} = \frac{4}{s(s+1/3)}$$

Since we are given

$$\mathcal{L}^{-1} \left\{ \frac{1}{s(s+a)} \right\} = \frac{1}{a} (1 - e^{-at}) u(t),$$

$$\mathcal{L}^{-1} \left\{ \frac{4}{s(s+1/3)} \right\} = 4 \left[\frac{1}{1/3} (1 - e^{-t/3}) u(t) \right]$$

$$\Rightarrow y(t) = 12(1 - e^{-t/3}) u(t)$$

$\Rightarrow$ Final long-term value:

$$\begin{aligned} & \lim_{t \rightarrow \infty} 12(1 - e^{-t/3}) u(t) \\ &= 12(1 - 0) \cdot 1 = 12 \end{aligned}$$

[2] An automated shuttle position loop is described by

$$\ddot{x}(t) + 4\dot{x}(t) + 13x(t) = 13r(t)$$

a) Determine the transfer function $G(s) = \frac{X(s)}{R(s)}$

steps: 1) set initial conditions to zero

2) Swap derivatives for s :

$$\Rightarrow s^2 X(s) + 4s X(s) + 13X(s) = 13R(s)$$

3) Rearrange

$$X(s)(s^2 + 4s + 13) = 13R(s)$$

$$G(s) = \frac{X(s)}{R(s)} = \frac{13}{s^2 + 4s + 13}$$

b) Compare the result with the standard 2nd-order form and determine ω_n and ζ .

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{13}{s^2 + 4s + 13}$$

$\Rightarrow$ Therefore:

$$\begin{aligned} \omega_n^2 &= 13 \\ \omega_n &= \sqrt{13} \frac{\text{rad}}{\text{s}} \end{aligned} \quad \checkmark$$

and

$$2\zeta\omega_n = 4$$

$$\zeta = \frac{4}{2\omega_n} = \frac{2}{\sqrt{13}} \quad \checkmark$$

c) Classify the system as overdamped, underdamped, or critically damped.

$$\zeta = \frac{2}{\sqrt{3}} \approx 0.55$$

$$0 < \zeta < 1 \quad \text{underdamped}$$

* keep table of damping ζ

[3] Two transfer functions are used for logistics equipment models.

a) For

$$G_1(s) = \frac{2}{s+5},$$

determine the order, the poles, the zeros, and the static gain $G_1(0)$.

Order: The order of the system is the highest power of s in the denominator

$G_1(s)$ is a 1st-order system!

Poles: The poles are the values of s for which the denominator becomes zero.

$$\boxed{\text{pole } p_1 = -5} \quad \checkmark$$

Zeros: The zeros are the values of s for which the numerator becomes zero.

$$\boxed{z_1 = \text{None}} \quad \checkmark \quad \text{No zeros!}$$

Static gain: The static gain is $G_1(0)$

$$G_1(0) = \frac{2}{0+5} = \frac{2}{5} = 0.4 \quad \checkmark$$

b) For

$$G_2(s) = \frac{s+2}{s^2+4s+8},$$

determine order, the zero, the poles, and the static gain $G_2(0)$.

⇒ Order: We see the highest power of s in the denominator is 2:

$$\Rightarrow \text{2nd-order} \quad \checkmark$$

zero: The zero

$$z_1 = -2 \quad \checkmark$$

Poles: the values of s for which $R(s) = 0$:

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-4 \pm \sqrt{16 - 32}}{2}$$

$$s = -\frac{4}{2} \pm \frac{4j}{2}$$

Poles:

$$p_1 = -2 + 2j$$

$$p_2 = -2 - 2j \quad \checkmark$$

Static gain:

$$G_2(0) = \frac{0+2}{0+0+8} = \frac{1}{4} \quad \checkmark$$

4 Consider the following three transfer functions:

$$G_A(s) = \frac{1}{(s+1)(s+3)},$$

$$G_B(s) = \frac{1}{s-2},$$

$$G_C(s) = \frac{5}{s^2 + 6s + 13}.$$

a) Determine the poles of all 3 systems.

$\Rightarrow$ Since the poles are roots of the denominator, we have:

Poles for $G_A(s)$: $p_1 = -1, p_2 = -3$ ✓

Poles for $G_B(s)$: $p_1 = 2$ ✓

Poles for $G_C(s)$:

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = \frac{-6 \pm \sqrt{36 - 52}}{2} = \frac{-6}{2} \pm \frac{4}{2}j$$

$$\Rightarrow \begin{aligned} p_1 &= -3 + 2j \\ p_2 &= -3 - 2j \end{aligned} \quad \checkmark$$

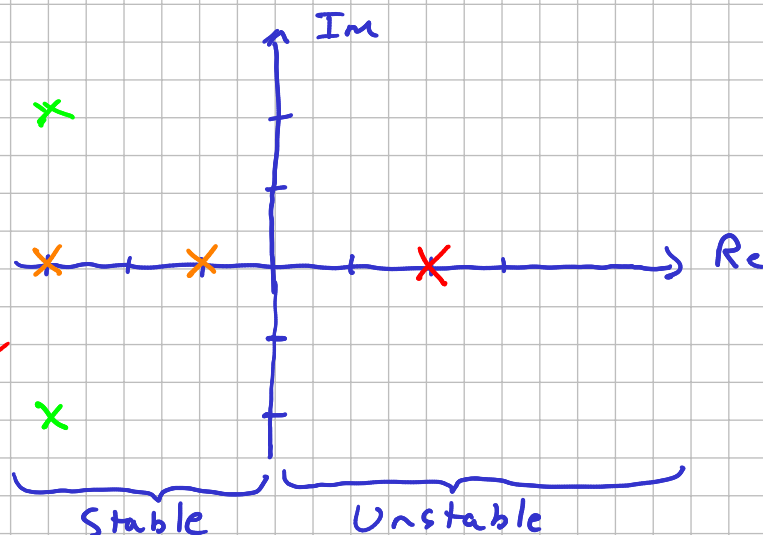
b) Classify each system as stable or unstable

$G_A(s)$

$G_B(s)$

$G_C(s)$

$G_B(s)$ is unstable ✓



$\Rightarrow G_A(s)$ and $G_C(s)$ are stable.

$G_B(s)$ is unstable.

Conceptual question: How could you stabilize the system described by B?

$\Rightarrow$ Add feedback!

$\Rightarrow$ choose a controller gain K that forces the poles into the stable region

c) Which system has oscillatory decay and why?

$\Rightarrow G_C(s)$

$\Rightarrow$ It has an oscillatory nature because it has complex poles.

$\Rightarrow$ It decays because the real components of its poles are negative!

9 A throughput regulation loop uses unity feedback with

$$G(s) = \frac{1}{3s+1}, \quad C(s) = 4$$

a) Determine the closed-loop transfer function

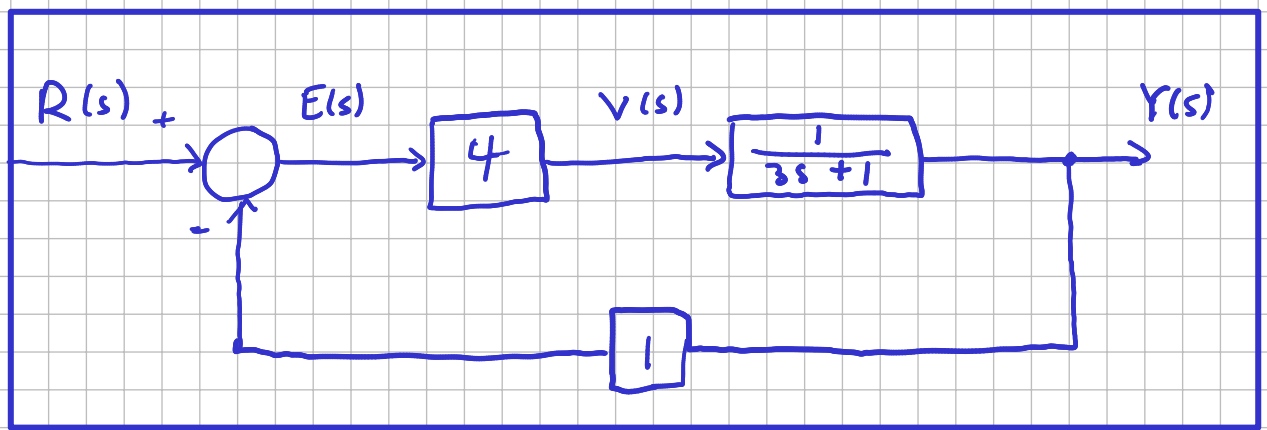
$$\frac{Y(s)}{R(s)}.$$

$\Rightarrow$ Let's identify our feedback control system components:

Plant: $G(s) = \frac{1}{3s+1}$

Controller: Proportional gain $C(s) = 4$

Unity Feedback: Sensor passes output through $H(s) = 1$



$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)C(s)}{1 + G(s)C(s)H(s)}$$

$$\Rightarrow G(s)C(s) = \frac{1}{3s+1} \cdot 4 = \frac{4}{3s+1}$$

$$\Rightarrow T(s) = \frac{\frac{4}{3s+1}}{1 + \frac{4}{3s+1}} \cdot \frac{3s+1}{3s+1}$$

$$T(s) = \frac{4}{(3s+1)+4} = \frac{4}{3s+5}$$

$$\Rightarrow T(s) = \frac{4/5}{\frac{3}{5}s + 1}$$

b) Determine the closed-loop pole and closed-loop τ

$\Rightarrow$ To get the pole, we set the denominator to zero:

$$\frac{3}{5}s + 1 = 0$$

$$\frac{3}{5}s = -1$$

$$3s = -5$$

$$s = -\frac{5}{3} = p \quad \checkmark$$

⇒ To get the time constant τ we look at the normalized transfer function:

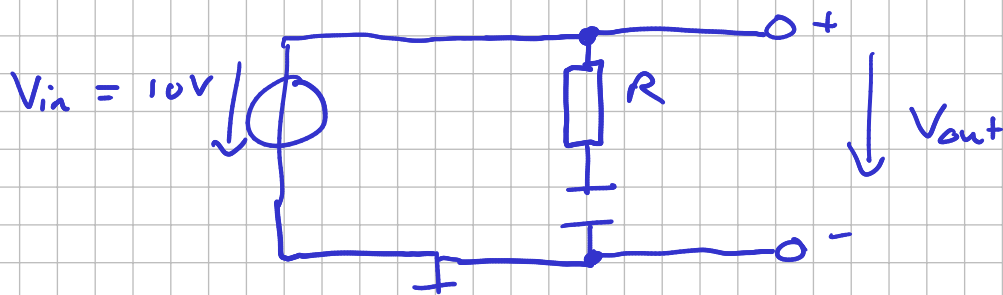
$$T(s) = \frac{4/5}{.3/5s + 1} = \frac{0.8}{\underbrace{0.6s + 1}_{\uparrow}}$$

$$\Rightarrow \tau = 0.6$$

⇒ The closed-loop τ_c is much smaller than the open-loop time constant τ_o . So the controller and feedback made the system faster!

C) Determine the DC gain and the steady-state error for a unit step reference.

⇒ The DC gain in electric circuit is the long-term value with constant input:



⇒ In other systems, this is the static gain given with $s = 0$.

$$T(0) = \frac{4}{3(0) + 5} = \frac{4}{5} = 0.8 \quad \checkmark$$

⇒ The steady-state error is the difference between what we input and what the system gives us at steady-state or DC Gain.

$$R(s) = 1$$

$$y_{ss} = 0.8$$

$$\Rightarrow 1 - 0.8 = 0.2 \quad \checkmark$$

20% error (steady-state).